

3.3 Metallic bonding homework — answers

CIE 9701 AS Chemistry · 8 multiple-choice + 11 written marks

Do not issue this file to students.

Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	A	A4	A	A7	B
A2	A	A5	A	A8	D
A3	A	A6	A		

Worked explanations

A1. A

2023 MJ Paper 11 Q2

Both calcium and barium are Group 2 metals with the same metallic structure, so the only difference that matters is the size of the cation. Ca^{2+} is smaller than Ba^{2+} , so the delocalised electrons are held closer to the Ca^{2+} ions and the electrostatic attraction is stronger, giving calcium the higher melting point.

B: metals do not have a “partly giant molecular” structure — they have a giant *metallic* lattice of cations in a sea of delocalised electrons.

C: both Ca and Ba contribute the *same* number (two) of delocalised electrons per atom; ionisation energy is not the factor here.

D: there is no such repulsion — the attraction between the cations and the delocalised electrons is what holds the lattice together.

A2. A

2021 MJ Paper 13 Q36

All three statements are correct and together explain the rise in melting point from Na to Al. Going $\text{Na} \rightarrow \text{Mg} \rightarrow \text{Al}$: (1) the charge on the metal ion increases (Na^+ , Mg^{2+} , Al^{3+}); (2) the number of delocalised electrons per ion increases (1, 2, 3); (3) the ionic radius decreases. A higher charge, more delocalised electrons and a smaller radius all strengthen the metallic bonding, so the melting point rises.

B / C: each drops one of the three correct statements.

D: keeps only statement 1 — the other two statements are also true.

A3. A

2024 MJ Paper 12 Q6

Boron in BF_3 has only **three** bond pairs and **no lone pair**, so there are only 6 electrons in its outer shell — an *incomplete* octet. This is why BF_3 is a good electron-pair acceptor (a Lewis acid).

B: in CH_3^- the carbon has 3 bond pairs *and* one lone pair = 8 outer electrons (complete octet).

C: in F_2O the oxygen has 2 bond pairs and 2 lone pairs = 8 electrons.

D: in H_3O^+ the oxygen has 3 bond pairs and 1 lone pair = 8 electrons.

A4. A

2021 FM Paper 12 Q34

All three species have a vacant orbital that can accept a lone pair, so all three can form a coordinate (dative covalent) bond. BF_3 : B has an empty 2p orbital. H^+ : the hydrogen ion has an empty 1s orbital. CH_3^+ : the carbon has an empty 2p orbital (only 6 outer electrons).

B / C / D: each option excludes at least one species that is in fact able to accept a lone pair. The trap is forgetting that H^+ and CH_3^+ , not just BF_3 , are electron-deficient.

A5. A

2024 MJ Paper 11 Q5

In the Al_2Cl_6 dimer the two aluminium atoms are joined through two *bridging* chlorine atoms. Each bridging Cl uses a lone pair to form a *coordinate (dative covalent)* bond to one Al, and the Al–Cl bonds are covalent. So Al_2Cl_6 contains **covalent and coordinate** bonding only.

B: “covalent only” ignores the dative bonds to the bridging Cl atoms — each Al is surrounded by 4 electron pairs, which needs the dative bonds.

C / D: Al_2Cl_6 is a molecular (covalent) species; it is not ionic. (It is AlCl_3 that behaves ionically in the solid state, not the Al_2Cl_6 dimer.)

A6. A

2023 ON Paper 11 Q5

Aluminium chloride *sublimes* to give a vapour of **Al_2Cl_6 dimer** molecules — two Al centres bridged by two Cl atoms, with four terminal Cl atoms. Option A is the dimer.

B / C: do not show the correct Al_2Cl_6 dimer arrangement (the vapour is not the simple AlCl_3 monomer at this temperature).

D: shows an ionic formulation, $\text{Al}^{3+}(\text{Cl}^-)_3$. The vapour consists of covalent Al_2Cl_6 molecules, not separate ions.

A7. B

2022 FM Paper 12 Q5

The correct dot-and-cross diagram must show each aluminium atom surrounded by **four** electron pairs (giving Al a complete octet) and must show the *dative* bond: a bridging chlorine atom donates a lone pair to an aluminium atom. The key on the paper distinguishes the electrons that came from Al from those that came from Cl. Diagram B is the one that shows the Al_2Cl_6 structure with the bridging Cl atoms donating a lone pair, and the correct number of electron pairs around each Al.

A: shows Al^{3+} and Cl^- ions with the wrong electron bookkeeping — Al_2Cl_6 is molecular and each Al still has four shared pairs.

C / D: show the wrong arrangement of atoms / electron pairs (they do not reproduce the bridged Al_2Cl_6 structure with the dative pairs correctly marked).

A8. D

2021 ON Paper 11 Q35

A chloride with *both* covalent and coordinate bonding must be a molecular chloride that forms a bridged dimer. (1) **Al** — forms Al_2Cl_6 , which has covalent bonds and dative bonds from the bridging Cl atoms. **Correct.** (2) **Si** — forms SiCl_4 , which has ordinary covalent bonds only (no dative bonding). **Wrong.** (3) **Mg** — forms MgCl_2 , which is ionic. **Wrong.** Only statement 1 is correct.

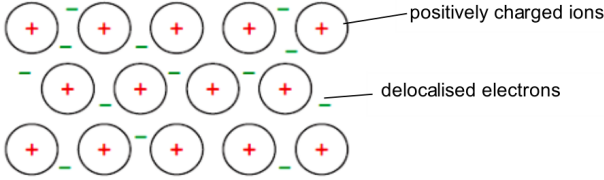
A / B / C: all include Si or Mg, whose chlorides do not contain coordinate bonding.

Section B — Mark scheme clips

B1(a) [2]

Question	Answer	Marks
2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	2

B2(c) [1]

1(c)	 positively charged ions delocalised electrons labelled diagram shows: - regular arrangement of minimum of 2×2 circles containing + for positive ions AND surrounded by (sea of) delocalised electrons	1
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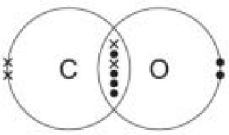
B3(a) [1]

Question	Answer	Marks
5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	1

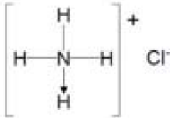
B4(b)(i) [1]

2(b)(i)	(covalent) bond with both electrons are provided from the same / one species OR shared pair (of electrons) are provided from the same species / one atom <i>owtte</i>	1
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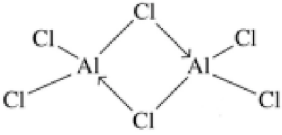
B4(b)(ii) [2]

2(b)(ii)	3 bonding pairs between C and O, 4 •'s AND 2×'s 1 lone pair on C, ××, AND 1 lone pair on O, ••. 	2
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B5(a) [2]

Question	Answer	Marks
4(a)	 OR Correct representation or label showing M1 (ionic bond between) NH_4^+ and Cl^- AND 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 M2 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons¹	2

B6(e)(i) [2]

2(e)(i)	 M1: 2 × coordinate bonds in the right place M2: all other bonds	2
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